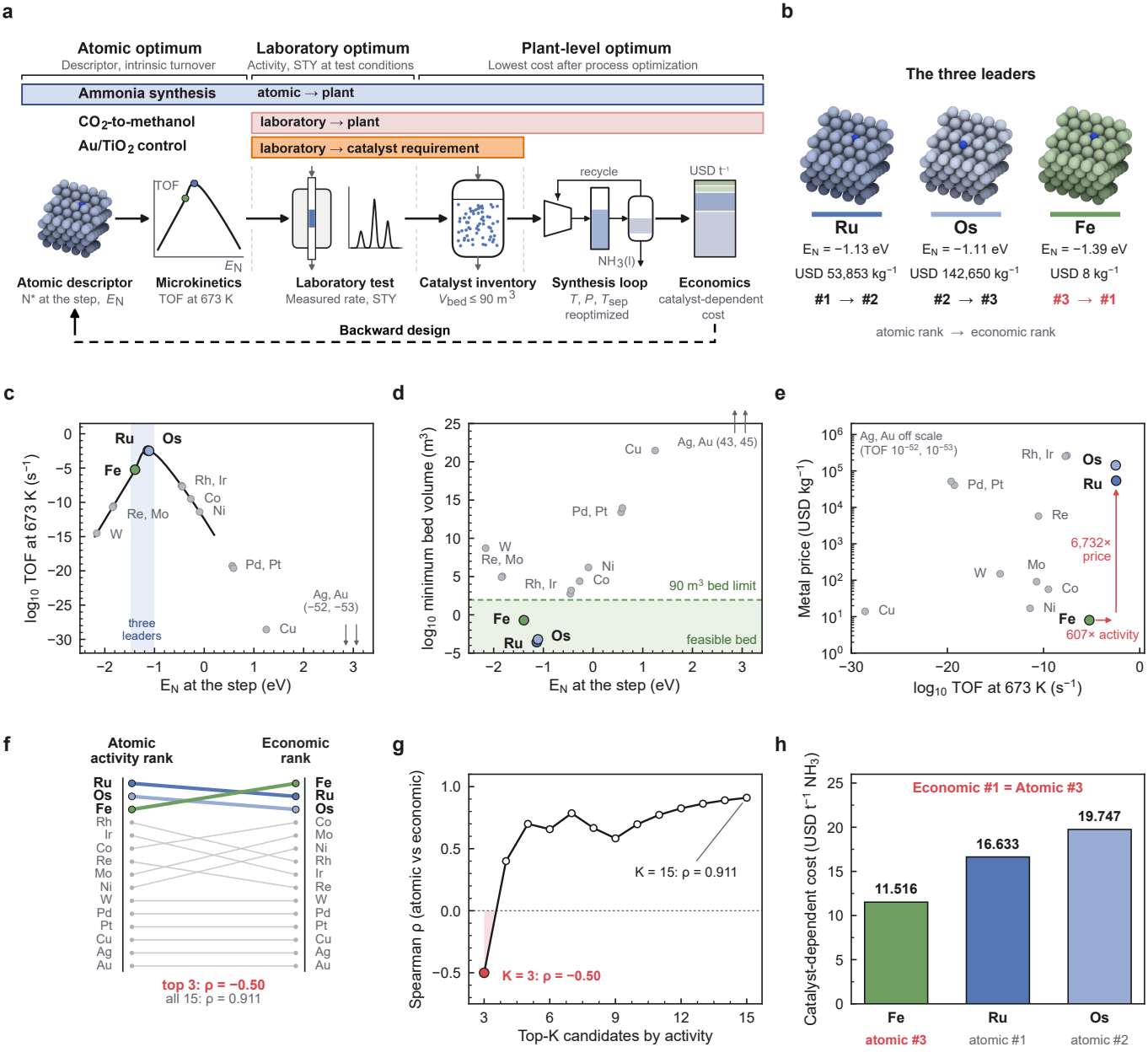

a
Atomic optimum
Descriptor, intrinsic turnover
Laboratory optimum
Activity, STY at test conditions
Plant-level optimum
Lowest cost after process optimization
Ammonia synthesis
atomic → plant
CO2-to-methanol
laboratory → plant
Au/TiO2 control
laboratory → catalyst requirement
TOF
EN
recycle
USD t⁻¹
NH3(l)
Atomic descriptor
N* at the step, EN
Microkinetics
TOF at 673 K
Laboratory test
Measured rate, STY
Catalyst inventory
Vbed ≤ 90 m3
Synthesis loop
T, P, Tsep
reoptimized
Economics
catalyst-dependent
cost
Backward design
b
The three leaders
Ru
EN = −1.13 eV
USD 53,853 kg−1
#1 → #2
Os
EN = −1.11 eV
USD 142,650 kg−1
#2 → #3
Fe
EN = −1.39 eV
USD 8 kg−1
#3 → #1
atomic rank → economic rank
c
log10 TOF at 673 K (s−1)
EN at the step (eV)
Ru
Os
Fe
Rh, Ir
Co
Ni
Re, Mo
W
Pd, Pt
Ag, Au
(−52, −53)
Cu
three
leaders
d
log10 minimum bed volume (m3)
EN at the step (eV)
Cu
Ag, Au (43, 45)
Pd, Pt
W
Re, Mo
Ni
Rh, Ir
Co
90 m3 bed limit
Fe
Ru
Os
feasible bed
e
Metal price (USD kg−1)
log10 TOF at 673 K (s−1)
Ag, Au off scale
(TOF 10−52, 10−53)
Rh, Ir
Os
Ru
Pd, Pt
Re
6,732×
price
W
Mo
Co
Cu
Ni
Fe
607× activity
f
Atomic
activity rank
Economic
rank
Ru
Os
Fe
Rh
Ir
Co
Re
Mo
Ni
W
Pd
Pt
Cu
Ag
Au
Fe
Ru
Os
Co
Mo
Ni
Rh
Ir
Re
W
Pd
Pt
Cu
Ag
Au
top 3: ρ = −0.50
all 15: ρ = 0.911
g
Spearman ρ (atomic vs economic)
Top-K candidates by activity
K = 15: ρ = 0.911
K = 3: ρ = −0.50
h
Catalyst-dependent cost (USD t−1 NH3)
Economic #1 = Atomic #3
11.516
16.633
19.747
Fe
atomic #3
Ru
atomic #1
Os
atomic #2